

7. The Listener

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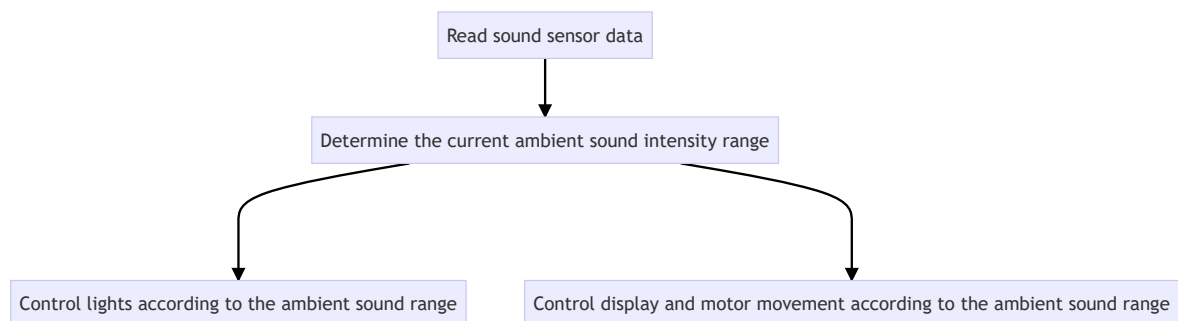
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Function Description

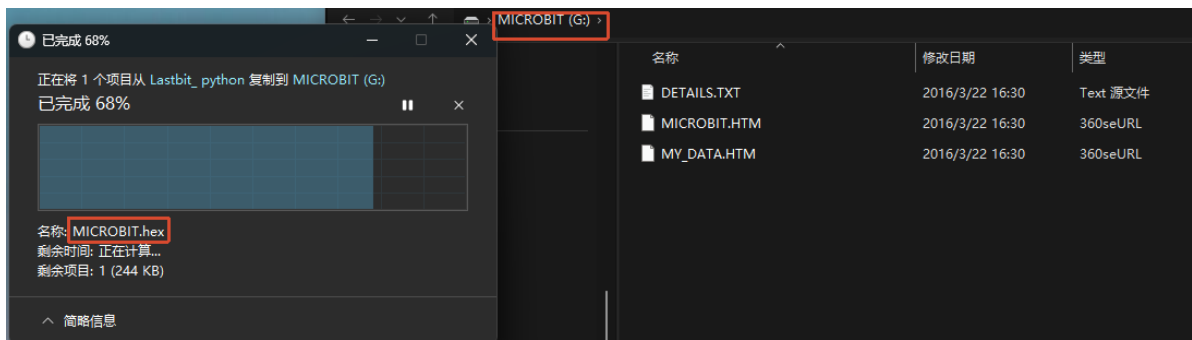
In this section, we will learn to use the microphone sensor to detect sound intensity, and control the car's rotation speed and LED flowing light effect according to the sound level, achieving the "Listener" effect.

Working Principle



Experiment Steps

Make sure `LASTBIT.hex` is flashed before flashing, otherwise an error will occur and the module cannot be imported



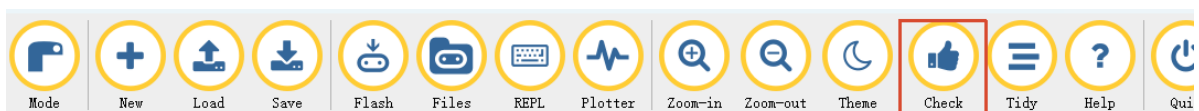
1. Before flashing, switch the switch to the OFF position and use a microUSB data cable to connect the computer and the micro:bit main board. **Note that it is the Microbit interface, not the Charging port on the expansion board.**
2. Open the Mu software. Here you can directly copy the program source code we provided; the operation steps are as follows. You can also enter the code manually in the editing window. Note! All English letters and symbols should be entered in English input mode. Use the Tab key for indentation, and the last line should end with a blank program.
Click the **New** button in the toolbar at the top left, and a file titled "untitled" will appear.



3. Copy the content of the **Code Implementation** section provided below into the current file. You will notice a red dot after the title bar, indicating that the code content has been modified but is in an unsaved state.

Whether to save or not does not affect code checking or flashing.

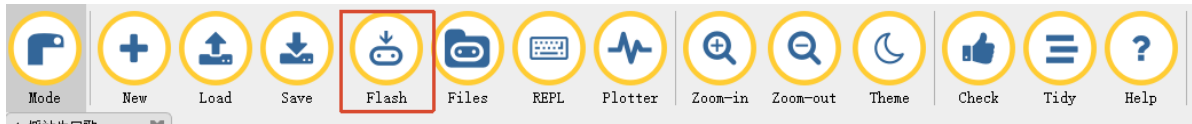
4. Now you can click **Check** to verify whether there are any problems with the code format. This case involves multi-line display patterns that may trigger warnings, but this does not affect flashing or running!



5. Now if you have already connected the MICROBIT to the PC in advance following the steps above, and flashed **LASTBIT.hex**, the next step is to download the current program to the Microbit.

Click the **Flash** tool in the toolbar to start flashing!

During flashing, the orange-yellow indicator light near the Microbit main board will blink frequently. When the editor shows `Copied code onto micro:bit.` at the bottom, it means the flashing is complete, and the indicator light will no longer blink after flashing.



6. Unplug the microUSB data cable and switch the switch to the ON position.

Code Implementation

Method 1: Use the built-in microphone of the micro:bit

```
# 7. V2 The Listener
from microbit import *
from lastbit import LASTBIT

Lastbit = LASTBIT()

# Delay time
delay = 300

while True:
    # Read sound level
    sound_level = microphone.sound_level()

    # Sound detection threshold can be adjusted

    # Display and move
    if sound_level > 110 and sound_level < 120:
        display.show(Image("00000:"
                           "00000:"
                           "00000:"
                           "00000:"
                           "99999"))
        Lastbit.car_control('SPINLEFT', 40, delay)
        Lastbit.stop_motor()
        Lastbit.car_control('SPINRIGHT', 40, delay)
        Lastbit.stop_motor()
        Lastbit.all_lights_running_water(250, 0, 0, delay)
    elif sound_level >= 120 and sound_level < 140:
        display.show(Image("00000:"
                           "00000:"
                           "00000:"
                           "09990:"
                           "99999"))
        Lastbit.car_control('SPINLEFT', 70, delay)
        Lastbit.stop_motor()
        Lastbit.car_control('SPINRIGHT', 70, delay)
        Lastbit.stop_motor()
        Lastbit.all_lights_running_water(0, 250, 0, delay)
    elif sound_level >= 140 and sound_level < 170:
        display.show(Image("00000:"
                           "00000:"
                           "00900:"
                           "99999:"
```

```

        "99999"))
    Lastbit.car_control('SPINLEFT', 110, delay)
    Lastbit.stop_motor()
    Lastbit.car_control('SPINRIGHT', 110, delay)
    Lastbit.stop_motor()
    Lastbit.all_lights_running_water(250, 250, 0, delay)
elif sound_level >= 170 and sound_level < 200:
    display.show(Image("00000:"
        "00000:"
        "99999:"
        "99999:"
        "99999"))
    Lastbit.car_control('SPINLEFT', 140, delay)
    Lastbit.stop_motor()
    Lastbit.car_control('SPINRIGHT', 140, delay)
    Lastbit.stop_motor()
    Lastbit.all_lights_running_water(0, 250, 250, delay)
elif sound_level >= 200 and sound_level < 230:
    display.show(Image("00000:"
        "09990:"
        "99999:"
        "99999:"
        "99999"))
    Lastbit.car_control('SPINLEFT', 170, delay)
    Lastbit.stop_motor()
    Lastbit.car_control('SPINRIGHT', 170, delay)
    Lastbit.stop_motor()
    Lastbit.all_lights_running_water(0, 0, 250, delay)
elif sound_level >= 230 and sound_level < 240:
    display.show(Image("00900:"
        "99999:"
        "99999:"
        "99999:"
        "99999"))
    Lastbit.car_control('SPINLEFT', 200, delay)
    Lastbit.stop_motor()
    Lastbit.car_control('SPINRIGHT', 200, delay)
    Lastbit.stop_motor()
    Lastbit.all_lights_running_water(150, 0, 250, delay)
elif sound_level >= 240:
    display.show(Image("99999:"
        "99999:"
        "99999:"
        "99999:"
        "99999"))
    Lastbit.car_control('SPINLEFT', 220, delay)
    Lastbit.stop_motor()
    Lastbit.car_control('SPINRIGHT', 220, delay)
    Lastbit.stop_motor()
    Lastbit.all_lights_running_water(250, 250, 250, delay)
else:
    display.clear()

```

Method 2: Use the MIC on the expansion board

```
# 7. V2 The Listener (Expansion board MIC)
from microbit import *
from lastbit import LASTBIT

Lastbit = LASTBIT()

# Delay time
delay = 300

while True:
    # Read sound level (from expansion board MIC)
    sound_level = Lastbit.sound_detect()[2]

    # Sound detection threshold can be adjusted

    # Display and move
    if sound_level > 100 and sound_level < 120:
        display.show(Image("00000:"
                           "00000:"
                           "00000:"
                           "00000:"
                           "99999"))
        Lastbit.car_control('SPINLEFT', 40, delay)
        Lastbit.stop_motor()
        Lastbit.car_control('SPINRIGHT', 40, delay)
        Lastbit.stop_motor()
        Lastbit.all_lights_running_water(250, 0, 0, delay)
    elif sound_level >= 120 and sound_level < 140:
        display.show(Image("00000:"
                           "00000:"
                           "00000:"
                           "09990:"
                           "99999"))
        Lastbit.car_control('SPINLEFT', 70, delay)
        Lastbit.stop_motor()
        Lastbit.car_control('SPINRIGHT', 70, delay)
        Lastbit.stop_motor()
        Lastbit.all_lights_running_water(0, 250, 0, delay)
    elif sound_level >= 140 and sound_level < 170:
        display.show(Image("00000:"
                           "00000:"
                           "00900:"
                           "99999:"
                           "99999"))
        Lastbit.car_control('SPINLEFT', 110, delay)
        Lastbit.stop_motor()
        Lastbit.car_control('SPINRIGHT', 110, delay)
        Lastbit.stop_motor()
        Lastbit.all_lights_running_water(250, 250, 0, delay)
    elif sound_level >= 170 and sound_level < 200:
        display.show(Image("00000:"
                           "00000:"
                           "99999:"
                           "99999:"
                           "99999"))
```

```

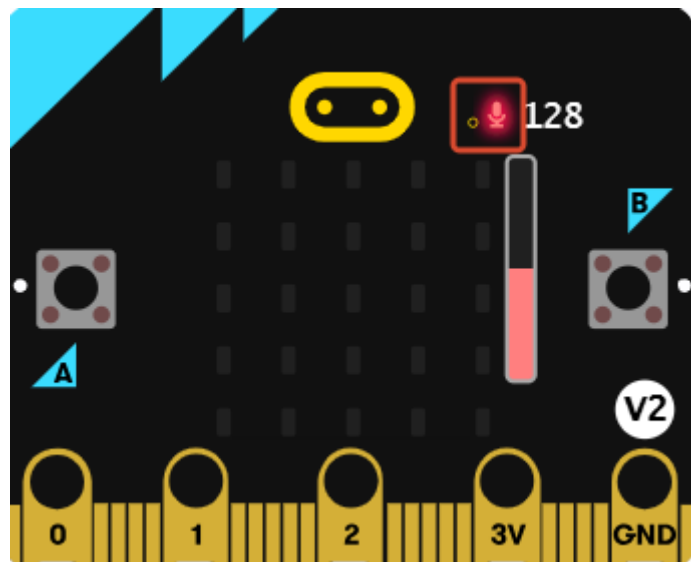
Lastbit.car_control('SPINLEFT', 140, delay)
Lastbit.stop_motor()
Lastbit.car_control('SPINRIGHT', 140, delay)
Lastbit.stop_motor()
Lastbit.all_lights_running_water(0, 250, 250, delay)
elif sound_level >= 200 and sound_level < 230:
    display.show(Image("00000:"
                        "09990:"
                        "99999:"
                        "99999:"
                        "99999"))
    Lastbit.car_control('SPINLEFT', 170, delay)
    Lastbit.stop_motor()
    Lastbit.car_control('SPINRIGHT', 170, delay)
    Lastbit.stop_motor()
    Lastbit.all_lights_running_water(0, 0, 250, delay)
elif sound_level >= 230 and sound_level < 240:
    display.show(Image("00900:"
                        "99999:"
                        "99999:"
                        "99999:"
                        "99999"))
    Lastbit.car_control('SPINLEFT', 200, delay)
    Lastbit.stop_motor()
    Lastbit.car_control('SPINRIGHT', 200, delay)
    Lastbit.stop_motor()
    Lastbit.all_lights_running_water(150, 0, 250, delay)
elif sound_level >= 240:
    display.show(Image("99999:"
                        "99999:"
                        "99999:"
                        "99999:"
                        "99999"))
    Lastbit.car_control('SPINLEFT', 220, delay)
    Lastbit.stop_motor()
    Lastbit.car_control('SPINRIGHT', 220, delay)
    Lastbit.stop_motor()
    Lastbit.all_lights_running_water(250, 250, 250, delay)
else:
    display.clear()

```

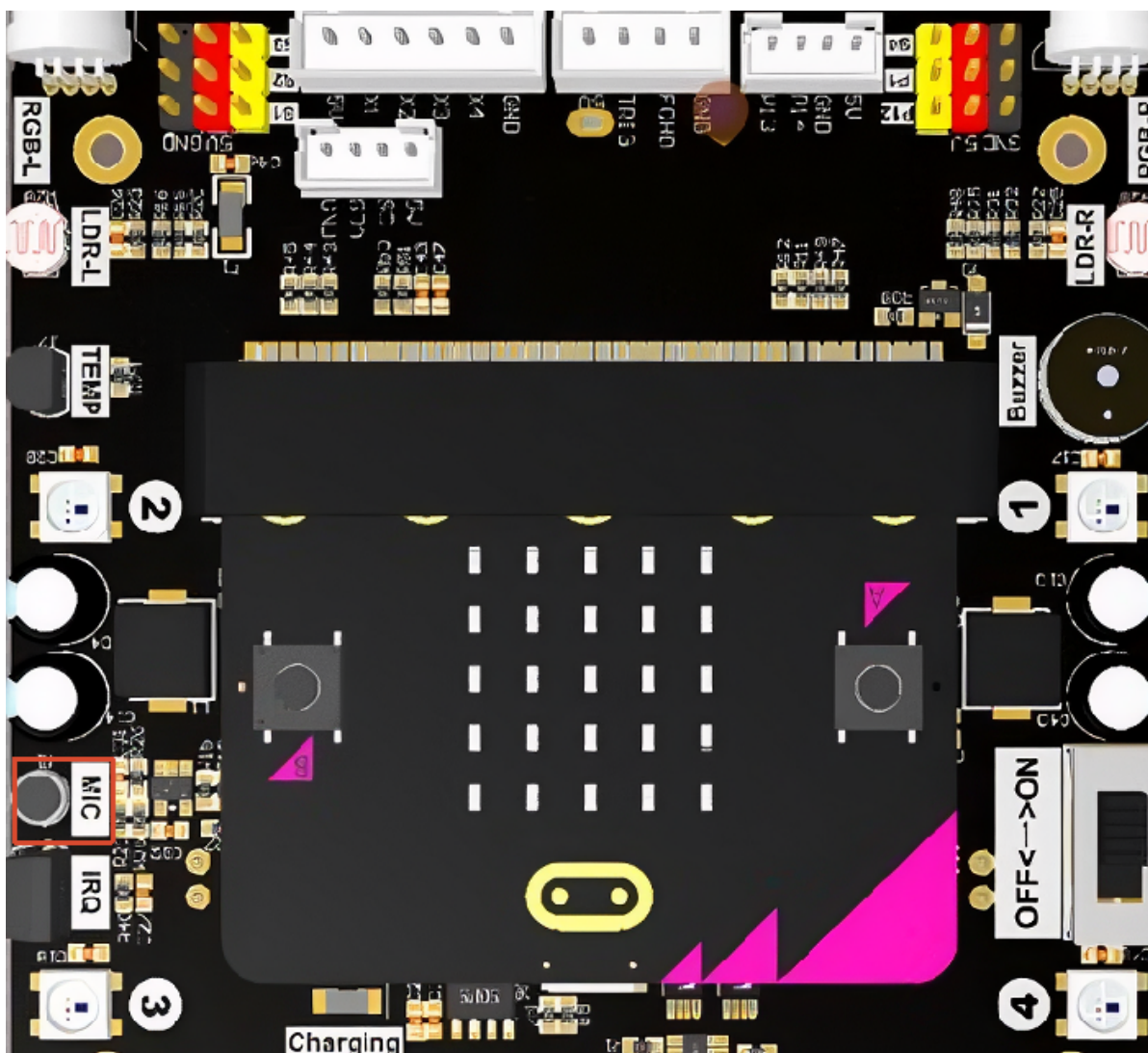
Experimental Phenomenon

The car will continuously detect the ambient sound. As the sound changes, the color of the flowing lights, the pattern displayed on the matrix screen, and the strength of the car's rotation will all change accordingly. The louder the sound, the larger the rotation angle of the car. Speaking close to the recording position will make it easier to detect the sound. You can also adjust the sound detection threshold according to the actual situation. If self-triggering occurs easily, you can raise the lower threshold.

The recording position of the built-in microphone of the micro:bit is as follows:



The recording position of the MIC on the expansion board is as follows:



Notes

- Ambient noise may affect the detection effect. It is recommended to conduct the experiment in a relatively quiet environment
- When using the built-in microphone of the micro:bit, use `microphone.sound_level()` to read the sound
- When using the MIC on the expansion board, use `Lastbit.sound_detect()[2]` to read the sound

- The sound detection threshold can be adjusted according to the actual environment